

Telco Customer Churn Data Pipeline & Scaling Report

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1. Overview

This document explains the data preparation, feature scaling, and partitioning workflow performed in Google Colab for the Telco Customer Churn project. The script cleans raw data, encodes categories, scales numerical features, and exports modular CSV files for team members.

2. Data Ingestion & Cleaning Code

```
import pandas as pd import numpy as np import seaborn as sns from matplotlib import
pyplot as plt df = pd.read_csv('/content/Dataset_ATS_v2.csv') df_clean = df.drop(columns=
['index'], errors='ignore') binary_cols = ['gender', 'Dependents', 'PhoneService',
'MultipleLines', 'Churn'] mapping = {'No': 0, 'Yes': 1, 'Female': 0, 'Male': 1} for col in
binary_cols: df_clean[col] = df_clean[col].map(mapping) df_clean = pd.get_dummies(df_clean,
columns=['InternetService', 'Contract'], drop_first=False, dtype=int)
```

3. Applied Scaling Technique

We used Scikit-Learn's StandardScaler to normalize continuous columns (tenure and MonthlyCharges) so they have a mean of 0 and standard deviation of 1.

```
from sklearn.preprocessing import StandardScaler scaler = StandardScaler()
df_clean[['tenure', 'MonthlyCharges']] = scaler.fit_transform(df_clean[['tenure',
'MonthlyCharges']])
```

4. Deduplication & Train-Test Split

Duplicate rows were removed, and the data was split into training and testing sets using an 80/20 stratified ratio.

```
duplicate_rows = df_clean.duplicated().sum() df_clean = df_clean.drop_duplicates()
df_clean.to_csv('cleaned_full_dataset.csv', index=False) X = df_clean.drop(columns=
['Churn']) y = df_clean['Churn'] from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42,
stratify=y ) X_train.to_csv('X_train.csv', index=False) X_test.to_csv('X_test.csv',
index=False) y_train.to_csv('y_train.csv', index=False) y_test.to_csv('y_test.csv',
index=False)
```

5. Deliverable Summary Table

File Name	Target User	Purpose
cleaned_full_dataset.csv	Aadesh (Clustering)	Full clean dataset
X_train.csv / X_test.csv	Sateesh (Prediction)	Feature matrices
y_train.csv / y_test.csv	Sateesh (Prediction)	Target labels

